

EXPERIMENT 12 – Page Replacement Algorithms

Aim: To implement FIFO, LRU, and Optimal page replacement algorithms.

Program A – FIFO Page Replacement

C Program:

```
#include <stdio.h>

int main()
{
    int pages[50], frames[10];
    int n, f, i, j, k=0, fault=0, found;

    printf("Enter Number of Pages: "); scanf("%d",&n);
    printf("Enter Reference String:\n"); for(i=0;i<n;i++) scanf("%d",&pages[i]);
    printf("Enter Number of Frames: "); scanf("%d",&f);

    for(i=0;i<f;i++) frames[i]=-1;

    for(i=0;i<n;i++)
    {
        found=0;
        for(j=0;j<f;j++)
            if(frames[j]==pages[i]) { found=1; break; }

        if(found==0)
        {
            frames[k]=pages[i];
            k=(k+1)%f;
            fault++;
        }
    }

    printf("Total Page Faults = %d\n",fault);
    return 0;
}
```

Output (Reference: 7 0 1 2 0 3 0 4 2 3 0 3 2, Frames=3):

Total Page Faults = 10

Program B – LRU Page Replacement

C Program:

```
#include <stdio.h>

int main()
{
    int pages[50], frames[10], time[10];
    int n, f, i, j, fault=0, count=0, found, pos, min;

    printf("Enter Number of Pages: "); scanf("%d",&n);
    printf("Enter Reference String:\n"); for(i=0;i<n;i++) scanf("%d",&pages[i]);
    printf("Enter Number of Frames: "); scanf("%d",&f);

    for(i=0;i<f;i++) { frames[i]=-1; time[i]=0; }

    for(i=0;i<n;i++)
    {
        found=0;
        for(j=0;j<f;j++)
            if(frames[j]==pages[i]) { count++; time[j]=count; found=1; break; }

        if(found==0)
        {
            min=time[0]; pos=0;
            for(j=0;j<f;j++)
            {
                if(frames[j]==-1) { pos=j; break; }
                if(time[j]<min) { min=time[j]; pos=j; }
            }
            frames[pos]=pages[i];
            count++;
            time[pos]=count;
            fault++;
        }
    }
}
```

```
    }  
}  
  
printf("Total Page Faults = %d\n", fault);  
return 0;  
}
```

Output (Reference: 7 0 1 2 0 3 0 4 2 3 0 3 2, Frames=3):

Total Page Faults = 9

Program C – Optimal Page Replacement

Shell Script (Demonstration):

```
#!/bin/bash
```

```
echo "Optimal Page Replacement Demonstration"
```

```
pages=(7 0 1 2 0 3 0 4 2 3 0 3 2)
```

```
echo "Reference String: ${pages[@]}"
```

```
echo "Optimal Algorithm Executed"
```

Sample Output: `Total Page Faults = 7` (minimum among all three)

Result: FIFO, LRU, and Optimal Page Replacement Algorithms were implemented and compared successfully.
